used in writing the test script for the counter smart contract. Listing 10.2 is the JS code for testing the counter smart contract. How do you develop the test code?

First, define the smart contract to be tested. Also declare that you are using the Truffle assertions framework and that a module is required. Then write the code to deploy and initialize the contract, using the `beforeEach()` function. Before execution of every `it()` test, the `beforeEach()` function is executed. Listing 10.2 provides a model test script to get you started.

The listing shows the independent tests (`it()`) for each function and each modifier in the smart contract. This simple test script consists of one `it()` function for each item to be tested. Each `it()` test has an `assert()` or `truffleAssert()` statement to check whether the test succeeded. This approach is simple, but a good starting point.

Listing 10.2 counterTest.js

```
const Counter = artifacts.require('../contracts/Counter.sol')                ❶
const truffleAssert = require('truffle-assertions');

contract('Counter', function () {
  let counter
  const negativeCounterError = 'Counter cannot become negative';
  const negativeValueError = 'Value must be greater than zero';

  beforeEach('Setup contract for each test', async function () {             ❷
    counter = await Counter.new()
    await counter.initialize(100)
  })

  it('Success on initialization of counter.', async function () {            ❸
    assert.equal(await counter.get(), 100)
  })
```